

# Measuring Incremental ROI of Sponsored Search Advertising

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Difference-in-Differences Analysis

## Executive Summary

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Bazaar.com aims to evaluate the true return on investment (ROI) of its sponsored search advertising on branded keywords. While initial calculations based on paid clicks suggest a high ROI, this approach does not distinguish between total traffic and incremental traffic, potentially overstating the value of advertising.

To address this issue, we leverage a natural experiment created by the temporary suspension of Google sponsored ads in weeks 10–12. Using a difference-in-differences framework with other search platforms as a control group, we estimate the causal impact of sponsored ads on total traffic.

The analysis finds a substantial decline in Google traffic after sponsored ads were suspended. The baseline Difference-in-Differences estimate is –9,911 visits per week relative to the control platforms, but a formal pre-treatment trend test rejects strict parallel trends. The estimate is therefore presented as a design-dependent incrementality signal, supported by sensitivity checks rather than as a definitive causal effect.

Using this estimate, we revise Bob’s ROI calculation by focusing on incremental traffic rather than total paid clicks. The resulting ROI is a scenario-based decision aid built from estimated incremental visits rather than observed paid clicks. Its exact magnitude should be interpreted with caution because the dataset is scaled and the identifying assumptions are imperfect.

The evidence supports continued testing and selective investment rather than an unconditional budget recommendation. Bazaar should prioritize segments where ads are more likely to generate incremental traffic and validate future decisions through stronger platform or geographic holdouts.

## Business Context

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Bazaar.com is a large online retailer that uses paid search advertising on platforms such as Google and Bing. The company bids on branded keywords that include its own name, such as searches containing “Bazaar,” which typically indicate strong user intent to visit the website.

When users search for branded terms, they are shown both sponsored ads and organic search results linking to Bazaar.com. Because these users are already aware of the brand and actively searching for it, they are likely to visit the website even without the presence of paid ads.

This creates a key business challenge: it is unclear whether sponsored ads for branded keywords generate additional traffic or simply capture users who would have arrived through organic search anyway. Bob's analysis assumes that all paid clicks are incremental, while Myra questions this assumption and suggests that many of these visits would have occurred regardless of advertising.

This distinction is critical because it directly affects how the company measures the return on its advertising spend. If a large portion of paid traffic is not incremental, then the true value of sponsored search advertising may be significantly lower than initially estimated.

## Why Click-Based ROI Is Biased

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Bob's calculation is problematic because it confuses **observed paid clicks** with **incremental clicks caused by advertising**.

His approach assumes that:

1. every sponsored click represents a new visitor generated by the ad, and
2. without the ad, these users would not have visited Bazaar.com.

These assumptions are unlikely to hold, especially for branded keyword searches. Many users searching for Bazaar.com already have strong intent to visit the website. If the sponsored ad were removed, a large share of these users would likely click on the organic link instead.

As a result, the ad often changes the channel of arrival (paid vs. organic) rather than generating truly new traffic. This substitution effect means that not all paid clicks are incremental.

At a deeper level, Bob's calculation fails to consider the counterfactual scenario, meaning what would have happened in the absence of advertising. Because it ignores this, the analysis overestimates the true causal impact of ads.

Therefore, Bob is measuring the profitability of paid clicks, rather than the profitability of incremental traffic, leading to an overestimation of ROI.

## Treatment and Control Design

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The unit of observation in this dataset is a **platform-week observation**.

The treatment is **the temporary cessation of Google sponsored search advertising beginning in week 10 due to a technical glitch. Google is the treated unit** because its sponsored ads were interrupted during this period. In contrast, **Bing, Yahoo, and Ask serve as control units** because their sponsored ads continued to run, providing a counterfactual for what would have happened to Google traffic in the absence of the interruption.

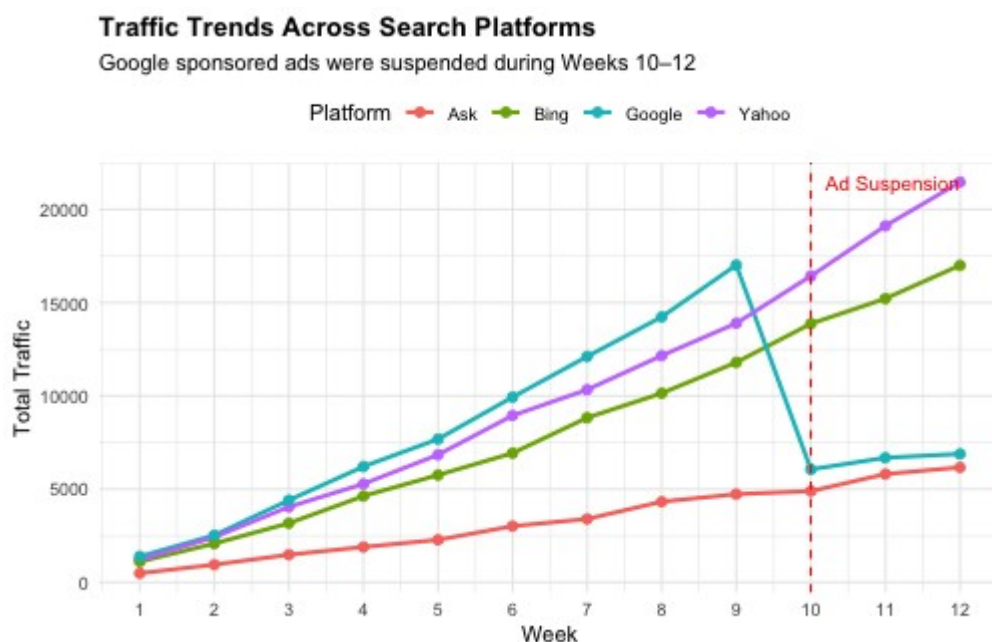
We define:

- after = 1 for weeks 10–12 and 0 for weeks 1–9
- treated = 1 for Google and 0 for Bing, Yahoo, and Ask

This setup forms a classic difference-in-differences design, where one unit experiences a treatment shock while comparable units do not. The treatment effect is captured by the interaction between the treated unit and the post-treatment period.

This design estimates an incremental effect by comparing the change in Google traffic before and after the interruption with the corresponding change in the control platforms. Its causal interpretation depends on the counterfactual assumptions evaluated later in the report.

## Overall Traffic Trends



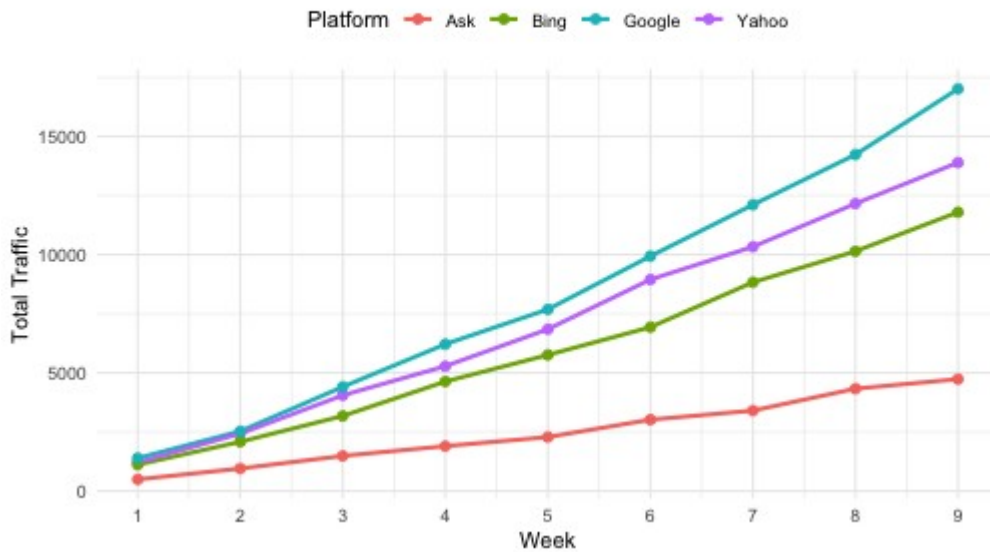
Before estimating the causal effect, we first examine traffic patterns across all search platforms over time. Weeks 10–12 correspond to the period during which Google sponsored ads were temporarily suspended because of a technical glitch.

Traffic across the platforms generally rises over time, while Google declines after the interruption and the controls continue upward. This descriptive pattern motivates a Difference-in-Differences analysis, but the figure alone cannot establish a causal effect or validate parallel trends.

## Parallel Trends Check

### Pre-Treatment Traffic Trends by Search Platform

Traffic trends before the Google sponsored ad suspension (Weeks 1–9)



Before applying the Difference-in-Differences model, we visually compare Weeks 1-9. All platforms trend upward, but visual similarity is not sufficient evidence of parallel trends. The formal slope test reported below detects a statistically significant Google-control trend difference, which weakens the strict causal interpretation of the baseline model.

## First-Difference Estimate

```
## [1] -1845.889
```

```
## [1] -0.2200135
```

### Results and Interpretation

The simple pre-post regression estimates the treatment effect as the coefficient on after, which is  $-1846$ . This indicates that average weekly Google traffic decreased by approximately 1,846 visits after week 10.

In percentage terms, average traffic declined from 8,390 to 6,544.00, corresponding to a 22.0% decrease.

### Why This Estimate Is Not Causal

Although this estimate suggests a substantial decline in traffic, it should not be interpreted as the causal effect of the ad interruption. The pre-post comparison assumes that the only change between the two periods is the removal of sponsored ads.

This assumption is unlikely to hold, as traffic may also be affected by time trends, seasonality, or other market-wide factors. As a result, the estimate may confound the true treatment effect with these external changes.

Therefore, the first-difference estimate is only a descriptive measure and does not provide a reliable causal estimate. To isolate the true effect of the ads, we need to compare this change to a control group, which motivates the use of a difference-in-differences approach.

## Difference-in-Differences Estimate

## [1] -9910.593

**Natural Experiment and Identification Assumptions** The temporary suspension of Google sponsored ads due to a technical glitch provides a natural experiment. Because the interruption was not driven by a strategic decision, it can be considered plausibly exogenous.

To interpret the difference-in-differences estimate as causal, several assumptions must hold. First, the parallel trends assumption requires that, in the absence of the treatment, Google traffic would have evolved in parallel with the control platforms. This implies that the change observed in the control group captures the underlying time trend. Second, the no interference assumption (SUTVA) requires that the treatment applied to Google does not affect traffic on the control platforms.

Under these assumptions, the control platforms provide a valid counterfactual for what would have happened to Google traffic without the treatment.

**Results and Interpretation** We estimate a Difference-in-Differences regression of total traffic on treatment status, time period, and their interaction. Under the identifying assumptions, the interaction coefficient represents the incremental effect of the suspension.

The baseline interaction estimate is **-9,910.59 visits per week** (conventional 95% CI: **-17,025 to -2,796**,  $p = 0.007$ ). Because the formal pre-treatment slope test rejects strict parallel trends, this is a baseline, design-dependent estimate rather than a definitive causal quantity.

**Comparison with the Pre-Post Estimate** The first-difference estimate suggested a decline of 1,846 visits per week, or about 22%. However, the difference-in-differences estimate is substantially larger in magnitude.

This difference indicates that the pre-post estimator fails to account for time trends that also affect the control platforms. As a result, it does not isolate the treatment effect from concurrent time trends.

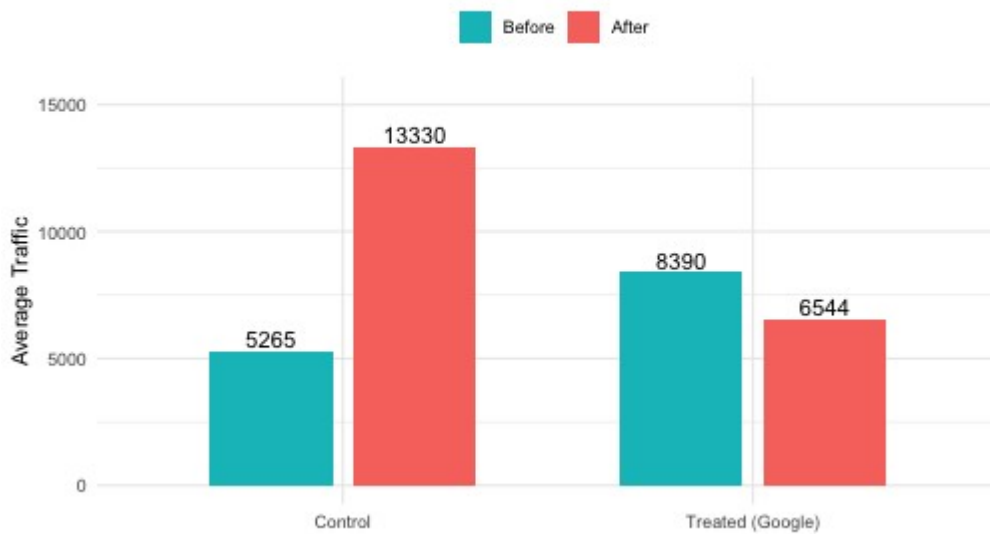
### Implications for the Post Estimator

The pre-post approach attributes all observed changes in Google traffic to the removal of sponsored ads, ignoring concurrent changes in the broader environment. In contrast, the difference-in-differences approach uses the control group to isolate the common time trend, providing a more credible estimate under the stated identification assumptions.

### Difference-in-Differences Visualization

## Difference-in-Differences Visualization

Comparison of treated and control groups before and after treatment



The figure compares the average traffic of the treated group (Google) and the control group before and after the suspension of sponsored ads. The control group shows a substantial increase in traffic over time, rising from 5,265 to 13,330, which reflects the underlying upward time trend in search activity.

In contrast, the treated group experiences a decline in traffic from 8,390 to 6,544 after the ads are suspended. This suggests that, while overall traffic in the market is increasing, Google traffic moves in the opposite direction during the same period.

The difference between these changes is the baseline DiD estimate. The treated group decreases while controls increase, producing a large negative interaction. This descriptive visualization is consistent with an incrementality effect, but the validity of the counterfactual remains limited by the pre-treatment trend difference.

## Uncertainty and Sensitivity Checks

The dataset contains only four platforms and twelve weeks, so statistical inference is exploratory. The confidence interval below uses the conventional OLS standard error from the specified DiD regression; it should not be interpreted as resolving the small-panel limitation.

| DiD Estimate | Std. Error | 95% CI Low | 95% CI High | p-value |
|--------------|------------|------------|-------------|---------|
| -9910.59     | 3530.03    | -17 024.89 | -2796.29    | 0.01    |

## Leave-One-Control-Out Sensitivity

This check re-estimates the DiD effect after removing Bing, Yahoo, and Ask one at a time. A wide range would indicate that the result is highly dependent on the selected counterfactual platform.

| Excluded Control | DiD Estimate | 95% CI Low | 95% CI High |
|------------------|--------------|------------|-------------|
| bing             | -9286.22     | -17 521.30 | -1051.15    |
| yahoo            | -8059.83     | -15 019.85 | -1099.81    |

| Excluded Control | DiD Estimate | 95% CI Low | 95% CI High |
|------------------|--------------|------------|-------------|
|------------------|--------------|------------|-------------|

|     |            |            |          |
|-----|------------|------------|----------|
| ask | -12 385.72 | -19 199.50 | -5571.94 |
|-----|------------|------------|----------|

### Formal Pre-Treatment Trend Check

A visual comparison can miss systematic slope differences. Using Weeks 1-9 only, the following model tests whether Google and the control platforms had different weekly trends before the ad suspension.

| Differential Weekly Trend | 95% CI Low | 95% CI High | p-value |
|---------------------------|------------|-------------|---------|
|---------------------------|------------|-------------|---------|

|        |        |         |      |
|--------|--------|---------|------|
| 786.28 | 129.49 | 1443.08 | 0.02 |
|--------|--------|---------|------|

The estimated pre-treatment slope difference is **+786 visits per week** (95% CI: **129 to 1,443**,  $p = 0.020$ ). This provides evidence against strict parallel trends, so the baseline DiD estimate should be treated as design-dependent rather than definitive.

### Group-Specific Linear-Trend Sensitivity

To assess whether the direction of the result survives a simple trend adjustment, we add a Google-specific linear trend to the DiD model.

| Trend-Adjusted Estimate | 95% CI Low | 95% CI High | p-value |
|-------------------------|------------|-------------|---------|
|-------------------------|------------|-------------|---------|

|            |            |          |   |
|------------|------------|----------|---|
| -14 252.08 | -21 941.48 | -6562.68 | 0 |
|------------|------------|----------|---|

The trend-adjusted estimate remains negative at **-14,252 visits per week** (95% CI: **-21,941 to -6,563**,  $p < 0.001$ ). The direction is stable, but the change in magnitude shows meaningful specification sensitivity.

### Pre-Treatment Placebo Check

A placebo interruption is assigned to Weeks 7-9 using only the actual pre-treatment period. A large placebo interaction would weaken the parallel-trends interpretation.

| Placebo Estimate | 95% CI Low | 95% CI High | p-value |
|------------------|------------|-------------|---------|
|------------------|------------|-------------|---------|

|         |          |         |      |
|---------|----------|---------|------|
| 3724.11 | -1200.63 | 8648.85 | 0.13 |
|---------|----------|---------|------|

These checks are diagnostics, not proof of identification. The preferred interpretation remains conditional on parallel trends, no spillovers across search engines, and no concurrent Google-specific shock.

## Understanding the Treatment Effect

| Metric | Estimate |
|--------|----------|
|--------|----------|

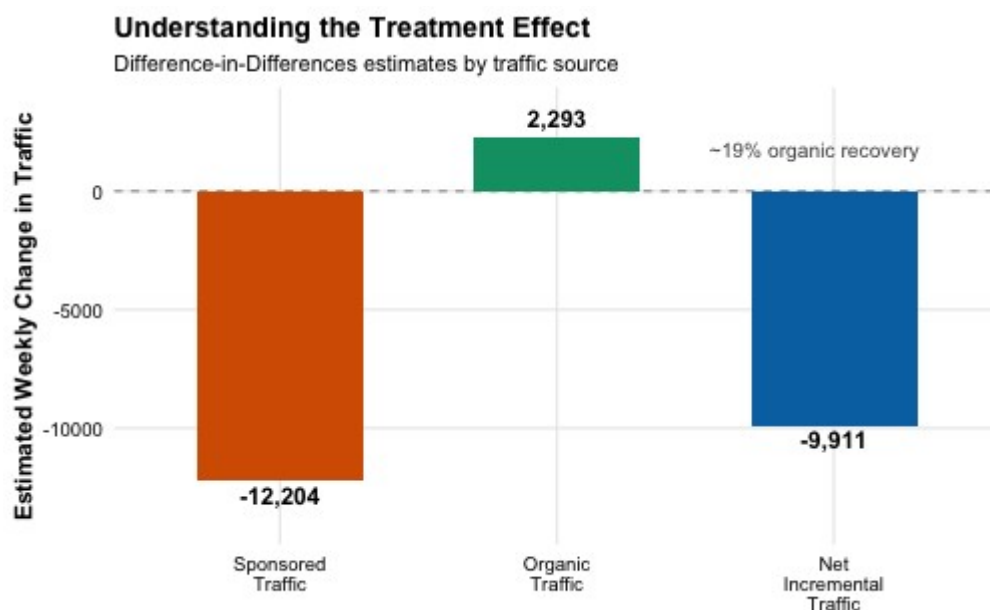
|                          |         |
|--------------------------|---------|
| Sponsored Traffic Effect | -12,204 |
|--------------------------|---------|

|                        |       |
|------------------------|-------|
| Organic Traffic Effect | 2,293 |
|------------------------|-------|

|                         |        |
|-------------------------|--------|
| Net Incremental Traffic | -9,911 |
|-------------------------|--------|

| Metric | Estimate |
|--------|----------|
|--------|----------|

Organic Recovery Rate    19%



The overall treatment effect can be decomposed into two offsetting components. When Google sponsored ads were suspended, sponsored traffic decreased by approximately 12,204 visits per week, while organic traffic increased by about 2,293 visits per week. This indicates that some users substituted from paid advertisements to organic search results instead of leaving the website entirely.

The estimated organic recovery rate is approximately 19%, meaning that about one-fifth of the lost sponsored traffic was recovered through organic search. Recovery did not fully offset the sponsored-traffic decline. This decomposition is consistent with positive incrementality, but the exact net effect remains sensitive to the counterfactual trend specification.

## ROI Correction Using Incremental Traffic

Bob's ROI is overstated because it assumes all paid clicks are incremental. In reality, many users would have visited through organic search even without ads. Therefore, we use the DiD estimate to measure incremental traffic. For financial translation, the baseline DiD point estimate uses 9,911 visits per week as a scenario input. Robustness specifications imply a wider traffic-effect range of approximately 8,060 to 14,252 visits per week, so the resulting ROI should be interpreted as a sensitivity scenario rather than an observed company return.

Using the DiD estimate, the corrected ROI is calculated as follows:

- Incremental visits: 9,911
- Conversion rate: 12%
- Incremental conversions: 1,189
- Contribution margin: \$24,975
- Advertising cost: \$3,674
- Estimated incremental ROI: **580%**



## Managerial Recommendations

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The analysis provides evidence consistent with positive incremental traffic: Google visits decline after ads are suspended relative to the comparison platforms. Because pre-treatment slopes differ, the exact causal magnitude is uncertain.

At the same time, the exact ROI magnitude should be interpreted with caution because the analysis is based on a scaled dataset. Therefore, the key takeaway is not the precise ROI value, but the direction and business implication of the effect: sponsored ads appear to generate positive incremental traffic, and future marketing decisions should be based on incremental impact rather than raw click volume.

Given these findings, Bazaar should continue investing in sponsored search advertising, but adopt a more disciplined and data-driven approach. In particular, the company should monitor incremental performance rather than relying on raw click-based metrics, and periodically validate results through controlled experiments where possible.

In addition, Bazaar may consider selectively optimizing its ad spend, focusing on segments or keywords where the incremental impact is more likely to be meaningful, rather than applying a uniform strategy across all branded searches.

Overall, the evidence supports continued testing and selective investment rather than an unconditional budget recommendation. Future geo or platform holdouts should be designed to create a stronger counterfactual and directly validate incremental ROI.